DIFFERENTIALS, DUAL ISOGENIES, AND THE TATE MODULE

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Here are three computations by hand, on the invariant differential, the dual isogeny and the Tate module. Each compares the multiplicative group $\mathbb{G}_{\mathrm{m}}$ with the elliptic curve

$$E\colon y^2 = x^3 - x, \qquad \omega_E \coloneqq \frac{\mathrm{d}x}{2y}.$$

Throughout, k is a field with $\operatorname{char} k \neq 2$, $\overline{k}$ an algebraic closure and $k^{\mathrm{sep}} \subseteq \overline{k}$ the separable closure. Here $\mathbb{G}_{\mathrm{m}} = \mathbb{P}^1_k \smallsetminus \{0, \infty\}$ has coordinate t and invariant differential $\omega_{\mathbb{G}_{\mathrm{m}}} \coloneqq \mathrm{d}t/t$. Each problem is preceded by the definitions and results it uses.

Reading. [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Proposition 3.1.

A formula. We use the following duplication formula on E, derived in Problem 1.1(5): if $x \in k(E)$ is the x-coordinate and $[2]\colon E \to E$ is multiplication by 2, then

$$[2]^*x \coloneqq x \circ [2] = \frac{(x^2+1)^2}{4(x^3-x)}. \tag{0.1}$$

1. Invariant differentials

Problem 1.1 (Poles and pullbacks)**.**
(1) Rewrite $\omega_{\mathbb{G}_{\mathrm{m}}}$ in the coordinate $s \coloneqq 1/t$ near ∞, and compute $\operatorname{div}(\omega_{\mathbb{G}_{\mathrm{m}}})$ on $\mathbb{P}^1_k$. Compute $[m]^*\omega_{\mathbb{G}_{\mathrm{m}}}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.
(2) Show that $\omega_E = \mathrm{d}y/(3x^2-1)$, and that y and $3x^2-1$ have no common zero on E. Deduce that ω_E has no zeros or poles on $E \smallsetminus \{O\}$.
(3) Using $\operatorname{ord}_O x = -2$ and $\operatorname{ord}_O y = -3$, show that $\operatorname{ord}_O \omega_E = 0$.
(4) Compute $[-1]^*\omega_E$ from $[-1](x,y) = (x,-y)$, and compare with $[-1]^*\omega_{\mathbb{G}_{\mathrm{m}}}$.
(5) We derive (0.1) by hand. Here $x, y \in k(E)$ are functions. For a point $P \in E(\overline{k})$ write $x_P \coloneqq x(P)$ and $y_P \coloneqq y(P)$ for its coordinates.
 (a) Let $P \in E(\overline{k})$ with $y_P \neq 0$. Show that the tangent line to E at P has slope $\lambda_P = (3x_P^2-1)/(2y_P)$, and use that its third point of intersection with E is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.
 (b) Let $\lambda \coloneqq (3x^2-1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.
(6) Solve this part in `Magma`, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function λ of (5), check (0.1), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3-x)}, \qquad g_0 = x^2+1, \quad g_1 = x^2-2x-1, \quad g_{-1} = x^2+2x-1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2+1)(x^4-6x^2+1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

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2. Dual isogenies

Consider the elliptic curves

$$E_1\colon y_1^2 = x_1^3 - x_1, \qquad E_2\colon y_2^2 = x_2^3 + 4x_2, \qquad E_3\colon y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i \coloneqq \mathrm{d}x_i/(2y_i)$ for the invariant differential of E_i. Consider the maps

$$\begin{aligned}
\varphi\colon E_1 &\to E_2, \quad (x_1, y_1) \mapsto \left(\frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2+1)}{x_1^2}\right),\\
\psi\colon E_2 &\to E_3, \quad (x_2, y_2) \mapsto \left(\frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2-4)}{x_2^2}\right),\\
\iota\colon E_3 &\to E_1, \quad (x_3, y_3) \mapsto \left(\frac{x_3}{4}, \frac{y_3}{8}\right).
\end{aligned}$$

The maps φ and ψ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies. Also ι is an isomorphism. That example computes the kernels and the dual, but not the constants c_α of Definition 2.1: that is part (2) of Problem 2.2.

Definition 2.1 (The constant c_α)**.** For an isogeny $\alpha\colon E_i \to E_j$, define c_α by $\alpha^*\omega_j = c_\alpha\,\omega_i$.

Problem 2.2 (A 2-isogeny and its dual)**.**
(1) Check that φ lands on E_2. Show that $\ker\varphi = \{O_1, (0,0)\}$, that $\varphi(\pm 1, 0) = (0,0)$, and that $\deg\varphi = 2$.[a]
(2) Compute c_φ.
(3) Check that ψ lands on E_3 and that ι lands on E_1, and compute c_ψ and c_ι.
(4) Show that $(\iota\circ\psi\circ\varphi)^* x_1 = [2]^* x_1$, so that $\iota\circ\psi\circ\varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat\varphi = \iota\circ\psi$, that $c_\varphi c_{\hat\varphi} = \deg\varphi$, and that $\ker\hat\varphi = \varphi(E_1[2](\overline{k}))$.
(5) Is there an isogeny $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2? If so, does it have a dual?

[a] *Hint:* write y_1 in terms of $\varphi^* y_2$ and x_1, and find a quadratic equation for x_1 over $\varphi^* k(E_2)$.

3. The Tate module

We first describe the Tate module of $\mathbb{G}_\mathrm{m}$, the familiar case, and then repeat each step for E. What links the two is the Weil pairing, which you may use without proof.

Proposition 3.1 (The Weil pairing [Sil09, Proposition III.8.1])**.** *Let $m \geqslant 1$ with $\operatorname{char} k \nmid m$. There is a pairing*

$$e_m\colon E[m](k^\mathrm{sep}) \times E[m](k^\mathrm{sep}) \to \mu_m(k^\mathrm{sep})$$

that is bilinear, alternating ($e_m(P,P) = 1$), non-degenerate, and Galois equivariant: $\sigma\big(e_m(P,Q)\big) = e_m(\sigma(P), \sigma(Q))$ for every $\sigma \in \mathrm{Gal}_k$. Here $E[m](k^\mathrm{sep}) = E[m](\overline{k})$, since $\operatorname{char} k \nmid m$. It follows that if (P,Q) is a basis of $E[m](k^\mathrm{sep})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P,Q)$ is a primitive m-th root of unity, and if σ acts on this basis by the matrix $A \in \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma\big(e_m(P,Q)\big) = e_m(P,Q)^{\det A}$.

Problem 3.2 (The 2-power torsion of $y^2 = x^3 - x$)**.**
Regard E as a curve over $\mathbb{Q}$, fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\overline{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\overline{\mathbb{Q}})$. Let σ be complex conjugation, acting on $\overline{\mathbb{Q}}$ and on $E(\overline{\mathbb{Q}})$ coordinatewise.
(1) Recall that $T_2(\mathbb{G}_\mathrm{m}) \coloneqq \varprojlim_n \mu_{2^n}(\overline{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$-basis of $T_2(\mathbb{G}_\mathrm{m})$, and write down its first three terms. How does σ act on $\mu_4(\overline{\mathbb{Q}})$, and on $T_2(\mathbb{G}_\mathrm{m})$?

(2) List $E[2](\overline{\mathbb{Q}})$. Using Problem 1.1(6), find the x-coordinates of all $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$, and conclude that $\#E[4](\overline{\mathbb{Q}}) = 16$.
(3) Show that $P_2 := (i, 1-i)$ and $Q_2 := (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\overline{\mathbb{Q}})$, with $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$. Show that (P_2, Q_2) is a basis of $E[4](\overline{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\overline{\mathbb{Q}})$ and Proposition 3.1.
(4) Show that σ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1,0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis (P_2, Q_2). Compare the determinant of the resulting matrix in $\mathrm{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of σ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Proposition 3.1.
(5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0,0)$, $Q_1 = (1,0)$ and P_2, Q_2 as above. Explain why they form a $\mathbb{Z}_2$-basis of $T_2(E)$, and compare with (1).

4. Magma examples

Example 4.1 (The isogenies of Problem 2.2)**.** We check in the function field of E_1 that φ and ψ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with Magma's own 2-isogeny.

```
E := EllipticCurve([-1,0]);                    // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;                          // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x);  y2 := y*(x^2 + 1)/x^2;              // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                                        // true
x3 := y2^2/x2^2;  y3 := y2*(x2^2 - 4)/x2^2;                 // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                                       // true
lam := (3*x^2 - 1)/(2*y);  xd := lam^2 - 2*x;  yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;                                  // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));                                 // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                                               // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)
```

Magma's dual is $\iota \circ \psi$ written out in the coordinates x_2, y_2 of E_2: $\big((x_2^2+4)/(4x_2),\ y_2(x_2^2-4)/(8x_2^2)\big)$.

Example 4.2 (Division polynomials and $E[4](\overline{\mathbb{Q}})$)**.** For $m \geqslant 2$, `DivisionPolynomial(E,m)` vanishes exactly at the x-coordinates of $E[m](\overline{\mathbb{Q}}) \smallsetminus \{O\}$. Its degree is $(m^2-1)/2$ for odd m and $(m^2+2)/2$ for even m, and its factors for $m = 4$ are x, $x \pm 1$ and the g_e of Problem 1.1(6).

```
[ Degree(DivisionPolynomial(E,m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E,4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8);  i := z^2;  s := z + z^7;  // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i];  Q2 := E8![1+s, 2+s];
2*P2, 2*Q2;                          // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4);            // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2;         // true: complex conjugation
```

Example 4.3 (Characteristic p)**.** Reducing modulo p, the p-division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

```
for p in [3,5,7,13] do
  Ep  := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
//  3 true   0  []
//  5 false 10  [ <x^2 + 2, 5> ]
//  7 true   0  []
// 13 false 78  [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]
```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$. In general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod 4$. For $p = 5, 13$ the division polynomial is the p-th power of a polynomial of degree $(p-1)/2$, whose roots are the x-coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$. The exponent p is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_{\mathrm{m}}$ is $t^p - 1 = (t-1)^p$.

Looking ahead. The Weil pairing of Proposition 3.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map φ_ℓ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]). Formal groups make c_φ an honest derivative. And Frobenius factors $[p]$ in characteristic p.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑